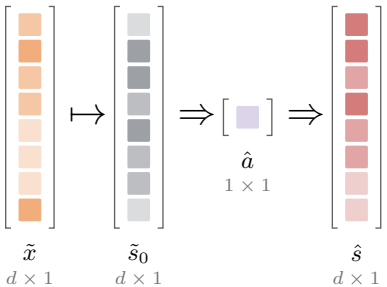
$$\hat{s} = \Pi\, x, \qquad \Pi = P\,(P^\dagger \Sigma^{-1} P)^{-1} P^\dagger \Sigma^{-1}, \qquad \min_P \frac{1}{N} \sum_i \|x_i - \Pi x_i\|^2_{\Sigma^{-1}}$$

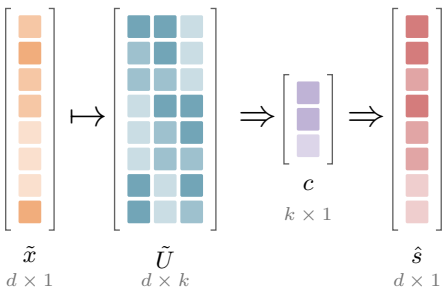
In whitened coordinates $\tilde{x} = \Sigma^{-1/2} x$ the operator Π is an ordinary orthogonal projection. The three rows minimize the same objective and differ only in the admissible column space of P.

Rank 1 — optimal filter: P fixed to a known template



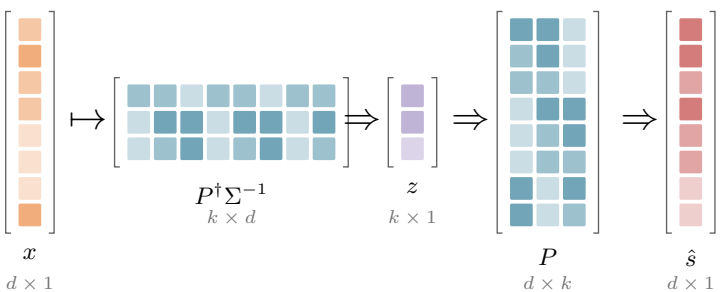
$\hat{a} = \tilde{s}_0^\dagger \tilde{x} \,/\, \tilde{s}_0^\dagger \tilde{s}_0$ — one scalar amplitude. The column space is the fixed line $\mathrm{span}(\tilde{s}_0)$, so no basis is estimated from data.

Rank k — EMPCA: P learned over the full Grassmannian $\mathrm{Gr}(k, \mathbb{C}^d)$



$c = \tilde{U}^\dagger \tilde{x}$, then $\hat{s} = \tilde{U} c$. The contraction runs over d: $(k \times d)(d \times 1) \to k \times 1$. The same d edge length is reused on both operands.

Tied linear autoencoder: the same subspace, written as encoder and decoder



Tie: $P^\dagger \Sigma^{-1} P = I_k$. The encoder is the Σ^{-1}-adjoint of the decoder, so their faces are physical transposes of one another.

Axes	d: observation dimension (time samples, or channels × samples after vectorization). k: retained rank, $k \ll d$. N: events.
Objects	$\Sigma \succ 0$: noise covariance, used as the metric rather than as a fitted parameter. $\tilde{s}_0$: fixed template. $\tilde{U}, P$: learned bases. $\hat{a}, c, z$: latent coefficients. $\hat{s}$: reconstruction.
Mechanism	Isotropic MSE replaces Σ^{-1} by I; it is the correct likelihood only when $\Sigma = \sigma^2 I$.

Σ^{-1}-orthogonal projection hierarchy: OF · EMPCA · tied linear autoencoder